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A dynamic weighted feature fusion lightweight algorithm for safety helmet detection based on YOLOv8

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ABSTRACT

To address the challenges encountered faced by existing current safety helmet detection algorithms in complex environments of construction site, such as excessive parameters, high computational demands, or deployment limitations on resource-constrained platforms, this paper proposes DWFL-YOLO, a dynamic weighted feature fusion lightweight network based on YOLOv8 for safety helmet detection. To start with, the diverse branch block (DBB) module is adopted to enhance the expressive power of a single convolution. Secondly, a novel dynamic weighted feature fusion network architecture is proposed to capture multi-scale information and improve model performance. After the feature fusion network is established, the partial convolutional gated linear units module is introduced to more accurately adjust the features from certain channels through a gating mechanism, ultimately achieving efficient feature fusion with less computation. Compared with YOLOv8n algorithm, the proposed algorithm has parameters reduced by 34.8 % and the Floating-point Operations FLOPs reduced by 14.8 % alongside 1.2 % increase in the value of Mean Average Precision (mAP). The achievement provides a new way of research in the lightweight aspect of the model.

1. Introduction

With the increasing attention paid to infrastructure construction, such as buildings, the safety issues faced by construction workers have becoming more prominent. The accident rate on construction sites in China remains high, largely due to inadequate safety measures and a lack of effective safety management. Compared to workers who do not wear safety helmets, those who do are less likely to suffer intracranial injuries in the event of a fall [1]. The use of helmets is considered one of the most important preventive strategies in construction. Monitoring the safety equipment and standardized behavior of construction personnel before and during their work can effectively reduce the occurrence of accidents.

Due to the large actual production site area, scattered worker trajectories, and limitations in power supply, data storage, and installation space. These conditions place higher demands on the accuracy and lightweight performance of safety helmet detection algorithms. Therefore, developing a high-performance and lightweight safety helmet detection algorithm based on computer vision to significantly reduce labor costs and effectively monitor the use of safety helmets on construction sites is of great significance.

Current mainstream safety helmet detection approaches can generally be classified into two categories: traditional algorithms based on handcrafted features and object detection methods leveraging deep learning. Rubaiyat et al. [2] were the first to integrate frequency domain information from images with human detection algorithms to detecting construction workers, utilizing color features and the circular Hough transform to assess whether individuals were wearing helmets. Liu et al. [3] focused on facial regions, employing a skin color detection technique and using Hu moments as feature vectors to recognize helmet usage. Additionally, Park et al. [4] applied Histogram of Oriented Gradients (HOG) features and Support Vector Machines (SVM) for the identification of helmets.

However, traditional algorithms extract low-level features, and require custom features for each object type, which makes it challenging to represent target characteristics effectively. Consequently, these methods often lack detection accuracy and robustness, and their inference speed is often slow.

As deep learning continues to advance, researchers have increasingly employed these techniques for helmet-wearing detection. Deep learning-based object detection methods are typically classified into two types: one-stage and two-stage algorithms. Two-stage methods are

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exemplified by the R-CNN series [5], whereas one-stage methods include the SSD [6] and YOLO series [7,8]. In particular, YOLOv5 [9] optimizes the backbone network structure to strike a balance between detection accuracy and speed, improving the overall performance in object detection tasks. Zhao et al. [10] proposed an efficient hybrid encoder, RT-DETR, which rapidly processes multi-scale features by decoupling intra-scale interactions and cross-scale fusion for end-to-end object detection to improve speed. Wang et al. [11] proposed YOLOv10, which introduces consistent dual assignment without NMS training, bringing competitive performance and low inference delay. Additionally, they introduced an efficiency-precision driven model design strategy for YOLO, comprehensively optimizing various components from the perspectives of efficiency and accuracy, greatly reducing computational costs and enhancing detection capabilities. Wu et al. [12] proposed an improved Faster R-CNN algorithm, which achieves the optimal network model through multi-scale feature fusion and candidate target box improvement, and applied it to helmet-wearing detection. Zhao et al. [13] proposed BDC-YOLOv5, an algorithm that significantly improves the ability to detect targets in complex scenes by introducing additional detection layers, BiFPN and CBAM modules. Li et al. [14] proposed a hierarchical positive sample selection (HPSS) mechanism during the training process and a post-processing algorithm based on box density, which improves the fitting ability of YOLOv5. Ding et al. [15] proposed the CM-YOLOv8s algorithm to detect safety helmets and their headbands, by employing the DeepSORT algorithm to track workers who were not wearing headbands.

Although some of the above object detection algorithms based on deep learning can improve the detection accuracy, these methods typically enhance detection accuracy by constructing deep and high-capacity neural networks. Therefore, the complexity and scale of these networks make it difficult to deploy them on platforms with fixed performance and limited computing resources, which is not suitable for helmet use detection tasks on construction sites.

Consequently, considering the complex environment and small detection targets of safety helmet detection, this paper proposes a lightweight and accurate deep learning-based method for safety helmet target detection. The main contributions of this article are as follows:

By introducing Diverse Branch Block (DBB) [16] structures with different receptive fields and complexities to replace the BottleNeck, which constructs the C2f-DBB module. The new module enriches the diversity of the feature space, improves the expressive ability of single convolution, and enhances the accuracy while maintaining the same number of parameters.

To better address multi-scale problems, a new dynamic weighted feature fusion structure is proposed. This structure utilizes bidirectional cross-scale connections to achieve simple and efficient multi-scale feature fusion, while employing a dynamic weight weighting mechanism for more efficient feature extraction.

To mitigate the relatively large parameter quantity of YOLOv8n, we introduced the PCGLU module, which is based on the Partial Convolution (PConv) [17] and Gated Linear Unit (GLU) [18]. The PCGLU module dynamically modulates channel weights based on localized features, enhancing the model's adaptability to complex backgrounds and varying environments. By integrating PConv, the model can reduce computational complexity while increasing sensitivity to critical channels, effectively improving robustness.

2. Related work

The challenge of building a lightweight object detection network remains a promising avenue for future research. In recent years, to facilitate the deployment of lightweight models across various application scenarios, numerous studies have proposed improved lightweight object detection algorithms. For instance, Guo Y et al. [19] designed a lightweight YOLO algorithm to optimize real-time performance on low-power edge devices. Additionally, to enable lightweight and low-power TinyML real-time vision processing systems, Xu K et al. [20] proposed a design combining an ultra-compact backbone network, a dedicated neural coprocessor (NCP), and a custom instruction set. Similarly, Chao D et al. [21] introduced a YOLOv5-lite lightweight model for detecting industrial infrastructure equipment, achieving an average precision of 94.4 % while significantly reducing model complexity, offering valuable insights for practical deployment in safety helmet detection. Typical lightweight networks include Slimneck [22]. EfficientViT [23], GhostHGNetV2 [24], and EfficientDet [25]. These networks, by optimizing their architectures and reducing parameters, offer new technical pathways for achieving efficient and reliable object detection in complex environments.

Against this backdrop, lightweight object detection algorithms have gradually been applied to the field of safety helmet detection to address the complexities of construction site environments and resource constraints. To meet this demand, numerous studies have proposed innovative safety helmet detection algorithms. For example, Lee et al. [26] proposed a safety helmet detection algorithm YOLO EfficientNet, which consists of two stages: first, using the YOLO model to detect objects corresponding to the human head, and second, using EfficientNet to classify the safety helmet wearing status of the detected object. Real-time performance is achieved with a limited amount of training data, ensuring the safety of construction personnel. Li et al. [27] proposed an YOLO-PL algorithm based on YOLOv4, which uses a DCSPX structure to expand the receptive field and designed a lightweight L-VoVN module with a one-shot aggregation architecture (OSA) and CSP structure. This design achieved a lightweight of the original continuous convolutional block. Chenet al. [28] proposed a lightweight facial assistance model for the detection of automatic welding cap usage based on an improved YOLOv5s. A Bidirectional Feature Pyramid Network (BiFPN) and Ghost module were introduced to reduce model complexity. Wang et al. [29] proposed a helmet-wearing detection model, YOLO-M, based on YOLOv5s. This model utilizes MobileNetv3 as the backbone network for feature extraction, thereby reducing the number of model parameters and overall model size. Additionally, residual edges are introduced in the feature fusion, enhancing the detection capability for small targets. Lin [30] proposed YOLOv8n SLIM-CA, based on the YOLOv8n algorithm. This model adopts a coordinate attention (CA) mechanism to enhance the focus on the safety helmet area in complex backgrounds, fuse features of different sizes extracted by the backbone network in the neck network, and add a small object detection layer in the detection layer. This approach reduces model complexity while maintaining accuracy. Han et al. [31] proposed a lightweight detection model based on FEV-YOLOv8n. This model introduced an EMA attention mechanism in the FasterC2f module to improve the backbone network by integrating spatial dependence and position information, thus improving detection accuracy. The VoVGSCSP module was also used to enhance the neck network, improving the recognition of occluded and small targets.

The above methods have made different lightweight improvements for helmet detection tasks. However, the basic models still have a high number of parameters and computation amount, leading to high computational demands in the lightweight detection models. This is not conducive to terminal deployment tasks and pays less attention to the performance in dense task detection and occlusion scenarios. In addition, most of the above methods use images with clean backgrounds and larger safety helmet targets. They also fail to take into account the multiple scale variations of the target. To detect small targets from different angles, considering the requirements of lightweight and accuracy, this paper adopts YOLO as the basic framework and develops DWFL-YOLO, which is a more efficient and accurate detection algorithm for small, multi-scale targets in helmet detection. This approach is more suitable for the complexity and variability of helmet detection application environments.

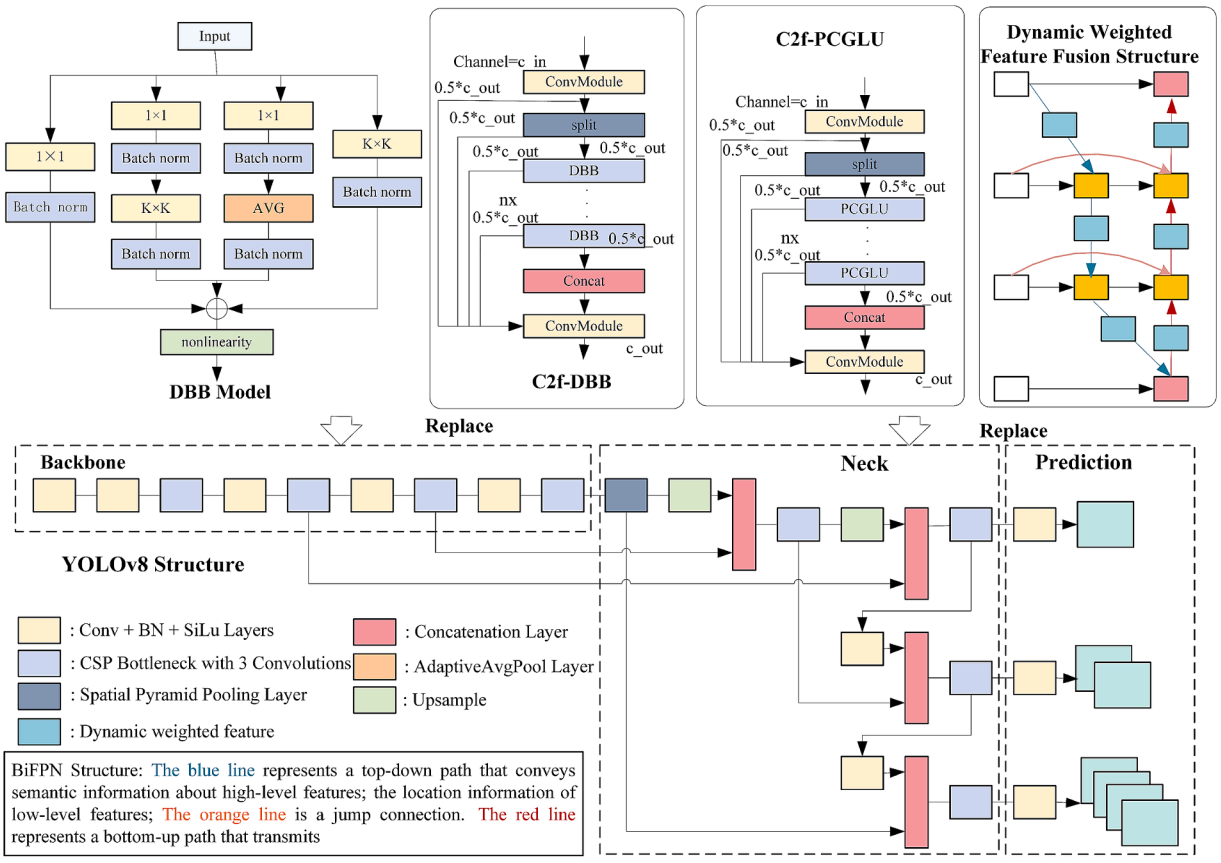


Fig. 1. Replace the BottleNeck in the original backbone network with the Diverse Branch Block (DBB) module to form C2f-DBB, and replace the C2f layer in the backbone network. Build a dynamic weighted feature fusion structure to replace the Neck network. And replace the C2f layer in the Neck network with the C2f-PCGLU module to further process the fused features.

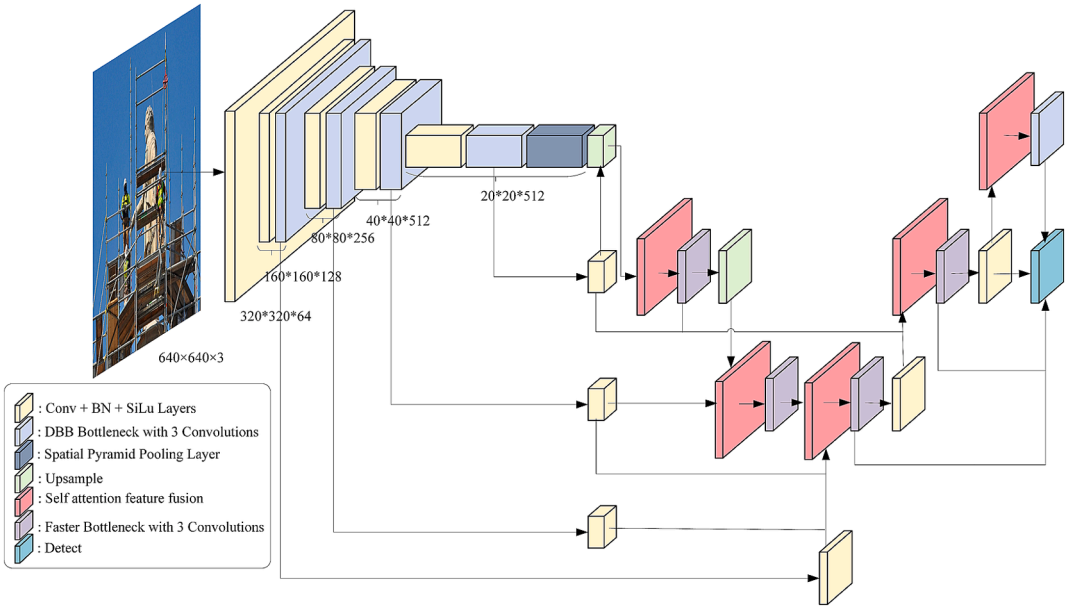


Fig. 2. DWFL-YOLO network structure.

3. Proposed method

3.1. Overall architecture

As shown in Fig. 1, this paper optimizes and improves the YOLOv8n network, resulting in a lightweight feature fusion network named DWFL-YOLO. This framework combines multi-scale feature fusion and lightweight networks to provide an overview of the proposed framework for helmet detection. The proposed network consists of three main components: (1) the DBB module, which expands the feature space by combining separate branches of different sizes and complexities, thereby enhancing the expressive power of individual convolutions. (2) A novel

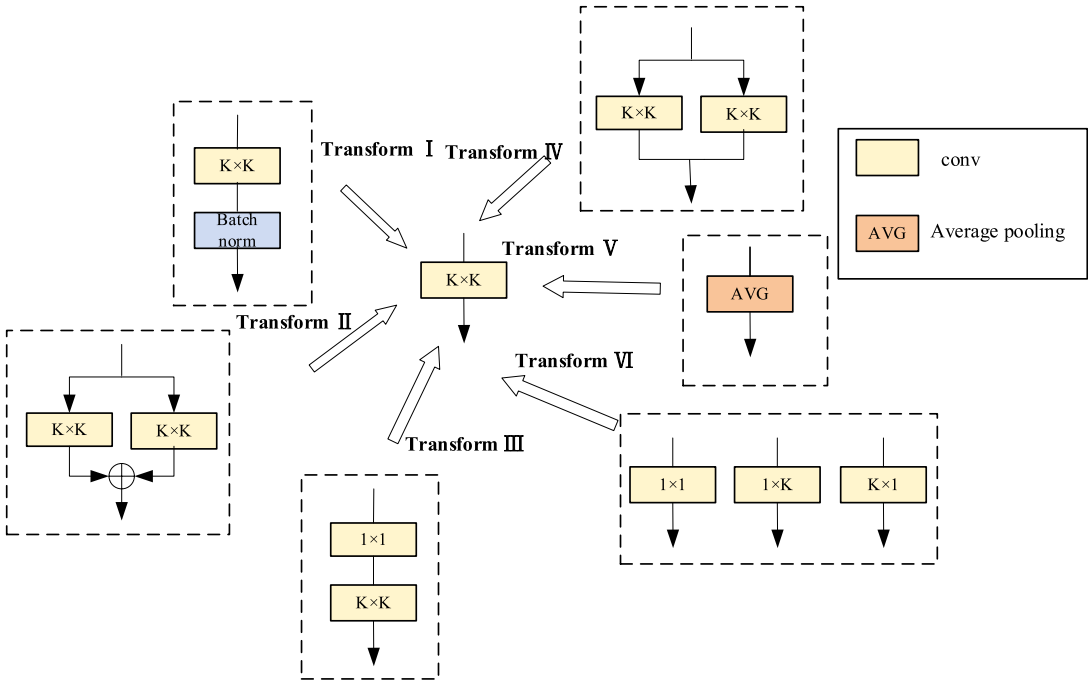


Fig. 3. (1) a conv for Conv-BN (2) a conv for branch addition (3) a conv for sequential convolutions (4) a conv for depth concatenation (5) a conv for average pooling (6) a conv for multi-scale convolutions.

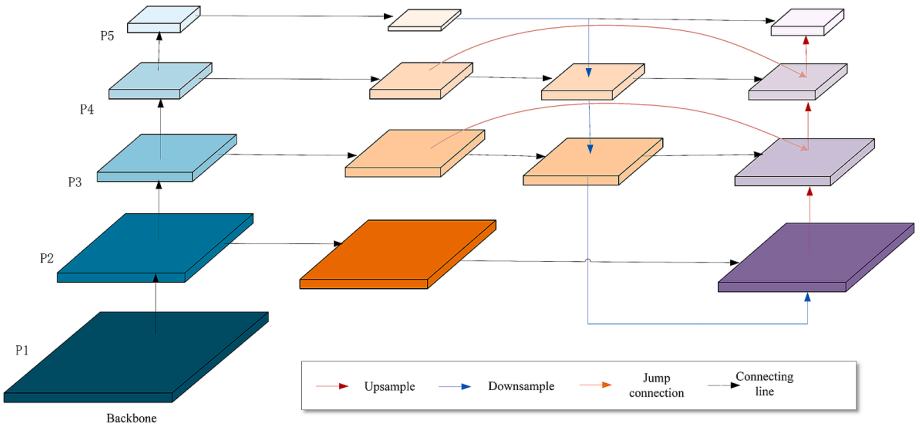


Fig. 4. Bidirectional feature pyramid network.

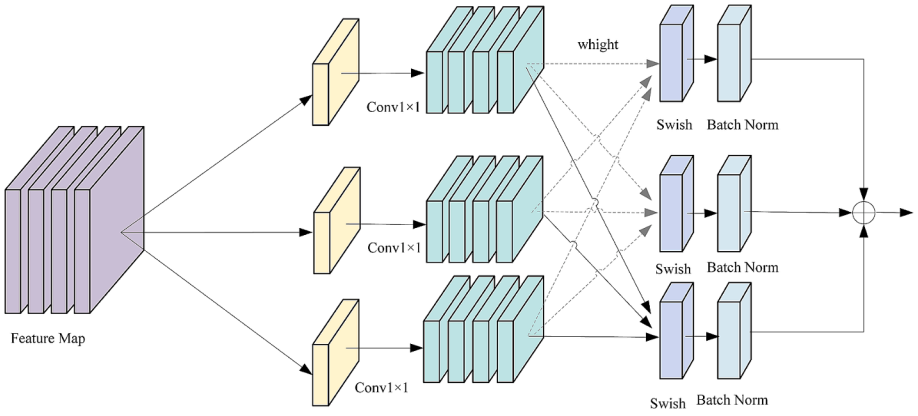


Fig. 5. Dynamic weighted feature. (1) Use dynamic weights to weight the input feature map. (2) By activating the function and normalizing the weights, ensure that their sum is 1. (3) Sum up the weighted feature maps to obtain the fused feature maps.

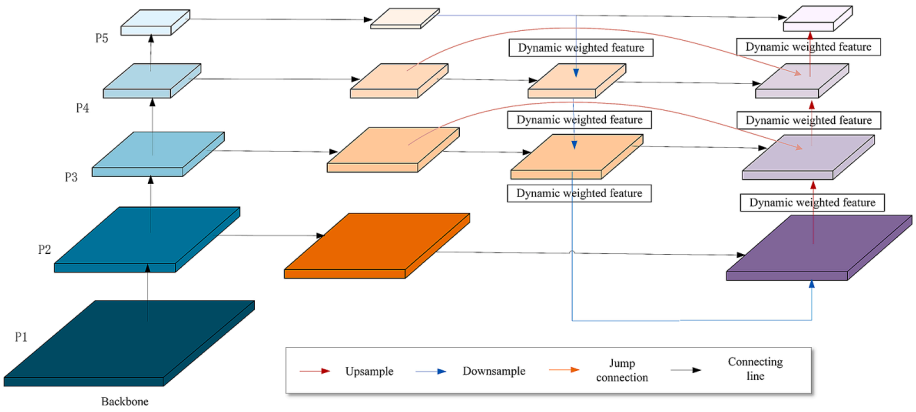


Fig. 6. Dynamic weighted feature fusion structure.

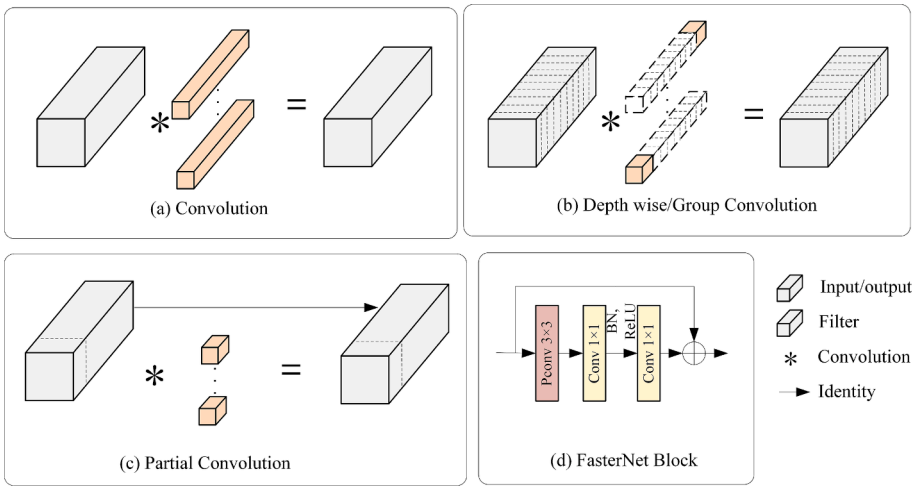


Fig. 7. (a) Standard convolution. (b) Depth wise Convolution (c) Partial Convolution performs spatial feature extraction on a few input channels, while keeping the remaining channels unchanged. (d) FasterNet Block.

dynamic weighted bidirectional feature fusion which improves the flexibility and effectiveness of feature fusion by adjusting the fusion weights of feature maps dynamically, thereby enhancing the performance and interpret ability of the model. (3) The PCGLU module not only enables precise capture of critical information in target regions through an optimized convolutional design but also reduces computational load and memory access requirements. The DWFL-YOLO network structure is shown in Fig. 2.

3.2. Diverse branch block

In safety helmet detection, the targets in images may appear in varying sizes and orientations, making multi-scale feature learning crucial for capturing the diverse features of helmets. The fusion method of C2f module in YOLOv8 mainly relies on feature concatenation, which has shortcomings in information selectivity. To overcome this limitation, this paper proposes a multi branch structure applied to backbone networks. Specifically, the DBB module is used to replace the Bottleneck block in C2f and improve the network's feature extraction capability through a multi branch structure. The new C2f module has been renamed as C2f-DBB module.

The DBB module is similar to the Inception module [32], combining convolutions of different scales to obtain receptive fields of varying sizes, enriching the diversity of the feature space, and enhancing the expressive power of a single convolution. Each branch uses convolution kernels of different sizes to capture features at different scales. This multi-branch design enables the network to effectively extract multi-scale features. On resource-constrained edge devices, the DBB module quantifies according to the computing power of the hardware to reduce the storage of model parameters and the computing load of model parameters.

Each branch within the DBB module can be configured differently, such as varying the number of convolutional layers, filter sizes, and activation functions. Small-scale feature branches can focus on fine-grained details, such as the spatial relationship and alignment between the helmet, ears, and face, while large-scale feature branches can capture the overall position and general shape of the helmet in the image. This approach enhances detection accuracy. The flexibility of the DBB module enables it to be adapted and optimized for specific tasks, improving adaptability and overall performance [33].

As shown in Fig. 3, the DBB module is ultimately transformed into a standard convolutional layer through a series of operations: fusing convolution with batch normalization (Transform I), merging convolutional layers with identical configurations (Transform II), combining sequential convolutions (Transform III), handling depth concatenation (Transform IV), converting average pooling into convolution (Transform V), and transforming multi-scale convolutions into standard convolution (Transform VI). This ensures that the inference process is equivalent to that of a model using a single convolutional layer, without introducing additional computational overhead or latency. This fusion creates richer and more diverse feature representations, enhancing the network's ability to recognize complex patterns. This flexibility allows the module

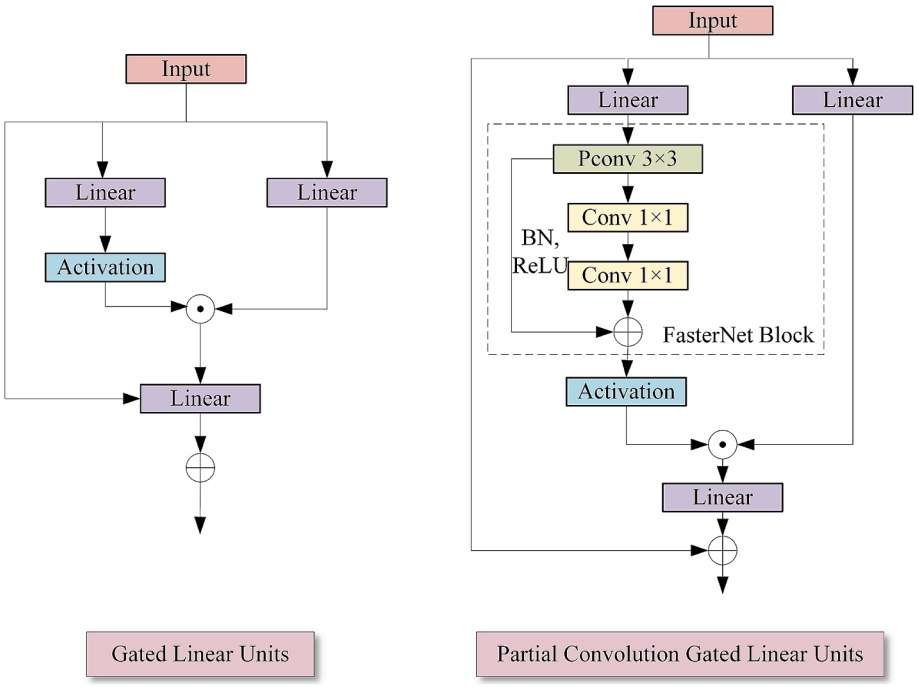


Fig. 8. Comparison of gated linear units and partial convolution GLU.

Table 1
Category statistics of the GDUT-HWD dataset.

Label	Blue	Yellow	White	Red	None
Train	2060	3114	2980	3485	3407
Val	552	784	801	746	964
Total	2612	3898	3781	4231	4371

to be adjusted and optimized for specific tasks, thereby improving adaptability and performance.

Through the six transformations mentioned above, the DBB block can be equivalently transformed into a convolution during the inference phase. Therefore, for any existing network such as ResNet [34], the 3x3 convolution can be replaced with a DBB module during the training phase, and then converted back during the inference phase to achieve lossless boosting.

3.3. *Dynamic weighted feature fusion structure*

The PAN-FPN (Path Aggregation Network, Feature Pyramid Network) structure of the Neck network in YOLOv8 is developed based on FPN. FPN mainly combines top-down and lateral connections to generate multi-scale feature maps for object detection [35]. On this basis, PAN-FPN has added bottom-up path aggregation, forming a bidirectional information flow. This structure enables rich detailed information from the bottom layer to be more effectively transmitted to the top layer, which is helpful for detecting small targets. At the same time, the semantic information from the top layer can be better integrated with the features from the bottom layer. Small targets occupy fewer pixels in the feature map, and during the PAN-FPN fusion process, alignment and fusion of features at different scales may result in deviations, leading to inaccurate localization of small targets. In complex construction sites, multiple workers work intensively, and inaccurate positioning of small target safety helmets may lead to misjudgment, resulting in adjacent targets being mistakenly or missed. The PAN-FPN structure involves a large number of convolutions, upsampling, and subsampling operations, resulting in high computational complexity. Running on embedded devices or low performance servers may not meet detection requirements due to insufficient resources. To address these issues, we propose a new module called Dynamic weighted feature fusion structure. The improved module we propose is based on BiFPN structure and introduces a dynamic fusion mechanism in the feature

Table 2
Experimental comparison results on the GDUT-HWD dataset.

Methods	AP/%					Params/M	FLOPs/G	Model size/MB	mAP (0.5)/%
	blue	white	yellow	red	none				
SSD	88.3	82.2	86.6	81.2	74.2	26.3	273.7	91.1	83.3
Faster-RCNN	92.2	84.4	87.8	84.0	80.2	133.2	148.2	521	85.8
RT-DETR	94.2	90.8	90.6	86.0	87.9	19.8	57.2	38.6	89.9
YOLOv5s	93.5	91.5	92.5	84.2	79.3	7.02	16.0	14.4	88.2
YOLOv5n	92.5	88.7	90.7	84.1	78.8	1.77	4.1	3.8	87.0
YOLOv7-Tiny	91.9	90.0	91.3	85.5	80.9	6.01	13.2	12.3	88.1
YOLOX-Tiny	84.9	80.7	81.9	80.3	75.2	5.06	10.4	36.9	80.6
YOLOv8s	95.0	91.1	90.9	86.3	88.0	5.06	10.4	36.9	80.6
YOLOv8n	94.1	90.7	90.5	85.9	87.8	3.01	8.1	6.2	89.8
YOLOv10n	93.9	92.5	90.6	87.3	86.5	2.70	8.2	5.8	90.1
YOLOv11s	95.1	92.0	92.2	86.9	87.1	9.41	21.3	19.2	90.7
YOLOv11n	94.3	91.4	91.0	86.0	87.4	2.58	6.3	5.5	90.0
DWFL-YOLOv8n(Ours)	95.5	91.4	90.9	88.1	88.7	1.96	6.9	6.1	90.9

Table 3
Experimental comparison results on the SHWD dataset.

Methods	AP/% hat	person	mAP (0.5)/%
SSD	83.8	73.1	78.5
Faster-RCNN	91.3	93.6	92.5
RT-DETR	89.2	94	91.6
YOLOv5s	95.1	92.9	93.7
YOLOv5n	93.7	87.3	91.0
YOLOv7-Tiny	88.2	91.5	89.8
YOLOX-Tiny	88.1	89.1	88.6
YOLOv8s	85.5	95.7	93.2
YOLOv8n	94.0	93.2	93.6
YOLOv10n	92.8	94.2	93.5
YOLOv11s	92.2	94.9	93.6
YOLOv11n	90.9	93.5	92.2
DWFL-YOLOv8n(Ours)	94.2	93.3	93.8

Table 4
Experimental comparison results on the CUMT-HelmeT dataset.

Methods	AP/% helmet	no-helmet	person	car	mAP (0.5)/%
SSD	78.3	74.5	90.6	98.9	85.6
Faster-RCNN	65.0	66.2	90.8	94.3	79.1
RT-DETR	86.5	87.0	89.5	90.0	88.3
YOLOv5s	88.2	81.1	94.5	99.5	90.8
YOLOv5n	92.4	80.8	90.1	99.5	88.2
YOLOv7-Tiny	76.9	70.6	89.1	93.9	82.6
YOLOX-Tiny	77.6	75.0	92.7	99.1	86.1
YOLOv8s	79.2	82.1	93.6	99.5	89.0
YOLOv8n	85.3	79.8	93.4	99.5	89.5
YOLOv10n	85.1	82.8	93.4	99.5	90.2
YOLOv11s	85.2	83.6	94.1	99.5	90.6
YOLOv11n	84.3	82.5	92.9	99.5	89.8
DWFL-YOLOv8n(Ours)	87.2	80.2	93.4	99.5	90.8

Table 5
Ablation experiments. We use the "DWFF" stand for dynamic weighted feature fusion structure.

Methods	C2f-DBB	DWFF	PCGLU	mAP (0.5)/%	Params/M	FLOPs/G
YOLOv8n				89.8	3.01	8.1
Methods (1)	√			90.2	3.01	8.1
Methods (2)		√		90.2	2.04	4.3
Methods (3)	√	√		90.5	2.04	7.5
Methods (4)		√	√	90.2	1.96	6.9
Methods (5)	√	√	√	90.9	1.96	6.9

fusion path to dynamically weight the feature map and enhance the expression of important features.

3.3.1. Bidirectional feature pyramid network

As shown in the Fig. 4, BiFPN incorporates both top-down and bottom-up bidirectional paths as feature network layers and repeats these layers multiple times [36]. This enables high-level semantic information and low-level detailed information to be repeatedly fused at different scales, allowing the network to learn richer and more discriminative feature representations. The approach is based on traditional feature pyramid networks, first optimizing cross-scale connections, removing single input edge nodes with small contributions, adding same-layer connections to fuse more features, and treating bidirectional paths as a single layer, repeating this process to enhance

Table 6
Comparison of state-of-the-art lightweight models.

Methods	Params/M	FLOPs/G	Model size/MB	mAP (0.5)/%
YOLOv8n-slimneck	2.80	7.3	5.9	89.4
YOLOv8n-efficientvit	4.01	9.4	8.7	88.3
YOLOv8-GhostHGNetV2	2.31	6.8	4.9	89.8
GCW-YOLOv8[46]	2.37	7.4	5.0	88.8
ShuffleNetV1[47]	16.6	37.4	33.6	89.4
MobileNetV1[48]	12.4	35	23.8	89.2
EfficientDet	6.8	6.1	–	87.8
ED-YOLO[49]	10.9	19.7	–	90.3
CIB-SE-YOLOv8[50]	2.7	7.6	–	88.6
[51]	5.46	12.6	11.5	89.1
DWFL-YOLOv8n(Ours)	1.96	6.9	6.1	90.9

the interaction between features of different scales. Weighted feature fusion is then employed to introduce learnable weights for input features of different resolutions. By using a fast normalization fusion method, the network learns the importance of features and overcomes the limitations of traditional simple addition [37].

The weighted feature fusion mechanism of BiFPN enables the network to learn the importance of different features. In helmet detection, it can assign higher weights to helmet features and suppress interference from background features. Through learning, the network will focus more on the color and shape of safety helmets. Reducing attention to features of cluttered background objects and more accurately extracting safety helmet features can improve detection accuracy.

3.3.2. Dynamic weighted feature network

However, the simple repetition of BiFPN layers primarily focuses on feature fusion within local levels, lacking an effective mechanism for integrating global information. In complex scene safety helmet detection, a semantic correlation exists between the target and the surrounding environment. The repetition of BiFPN layers hinders the integration of global contextual information into feature representation, thus impairing the accurate understanding and localization of the target. In the network architecture of YOLOv8, replace the PAN-FPN structure with an improved BiFPN module.

This paper has retained the core idea of bidirectional feature fusion that is repeated multiple times in BiFPN, while optimizing its connection method. To address this, this paper introduces a dynamic fusion mechanism in the inter-layer feature fusion operation. Firstly, a 1×1 convolution is applied to each input feature map to perform a linear transformation, altering its feature representation across channels while preserving the spatial dimensions. Next, the weights undergo a nonlinear transformation using the Swish activation function and are normalized to enhance the stability of the feature fusion process. Finally, the different feature maps are weighted and fused into a new feature map, which retains the most important information from the original input feature maps while accounting for the varying weight assignments across them.

As shown in Fig. 5, X_i is the i-th input feature map, Y_i is the feature map after applying 1x1 convolution, and w_i is the original attention weight of the i-th feature map. The activation function used is Swish [38], and ξ is a small constant to prevent division by zero errors. The fused feature map is denoted as F, as shown in Eqs. (1)–(6).

Input features undergo 1x1 convolution:

$$Y_i = \mathrm{Conv1} \times 1(X_i). \tag{1}$$

Activated attention weight:

$$a_i = w_i \cdot \sigma(w_i). \tag{2}$$

Normalization processing:

Table 7
Experimental results of DBB model on different datasets.

Dataset	YOLOv8n mAP (0.5)/%	mAP (0.5–0.95)/%	R	YOLOv8n-DBB mAP (0.5)/%	mAP (0.5–0.95)/%	R
GDUT-HWD	89.8	59.8	83.4	90.2	59.7	85.3
SHWD	93.6	60.5	88.3	93.5	60.1	88.0
CUMT-HelmeT	89.5	62.4	86.0	91.0	66.2	87.1
VOCdevkit2012	44.3	30.2	45.4	45.5	30.6	43.6

Table 8
Compared the results of BiFPN and DWFF.

Methods	Params/M	FLOPs/G	Model size/MB	mAP (0.5)/%	mAP (0.5–0.95)/%	FPS
YOLOv8n-BiFPN	1.99	4.2	7.1	90.2	60.2	185.2
YOLOv8n-DWFF	2.04	4.3	7.5	90.4	60.2	188.7

Table 9
Performance comparison of multiple activation functions.

Methods	mAP (0.5)/%	mAP (0.5–0.95)/%	FPS
DWFF + Swish	90.2	60.0	172.4
+ReLU[52]	90.0	59.8	161.3
+H_swish	90.3	59.8	163.9
+H_sigmoid[53]	90.4	60.2	166.7

Table 10
Compared the results of FasterNet and PCGLU.

Methods	Params/M	FLOPs/G	Model size/MB	mAP (0.5)/%	mAP (0.5–0.95)/%
DWFL-YOLOv8n (FasterNet)	1.97	7.0	6.1	90.2	60.0
DWFL-YOLOv8n (PCGLU)	1.96	6.9	6.1	90.9	60.4

$$\alpha_i = \frac{a_i}{\sum_{j=1}^{n} a_j + \xi}. \tag{3}$$

Weighted sum:

$$F = \sum_{i=1}^{n} \alpha_i Y_i. \tag{4}$$

The expression using p_4 as an example is:

$$P_4^{td} = Conv\left(\frac{w_1 \cdot P_4^{in} + w_2 \cdot \mathrm{Resize}(P_5^{in} + F_{5_in})}{w_1 + w_2 + \varepsilon}\right). \tag{5}$$

$$P_4^{out} = Conv\left(\frac{w_1' \cdot P_4^{in} + w_2' \cdot P_4^{td} + w_3' \cdot \mathrm{Resize}(P_3^{out} + F_{3_out})}{w_1' + w_2' + w_3' + \varepsilon}\right). \tag{6}$$

where *Resize* is usually a up sampling or down sampling op for resolution matching, and Conv is usually a convolutional op for feature processing. Where P_4^{td} is the intermediate feature at level 4 on the top-down pathway, and P_4^{out} is the output feature at level 4 on the bottom-up pathway.

In the model, this paper adopt the Swish activation function [39]. Owing to its superior performance compared to traditional activation functions. A key feature of Swish is its self-regulation mechanism, where larger input values result in greater output amplification, but this amplification slows as the values increase. This allows the Swish function to effectively suppress small activation values without overly diminishing large ones, helping the model automatically adjust the importance of each feature when processing feature maps.

In the new model, attention weights are non-linearly transformed using the Swish function. Unlike ReLU, Swish can retain negative value activation, which is crucial for maintaining the expressive capability of negative values in certain tasks. By utilizing the Swish function, the adjustment of attention weights becomes more flexible, which better adapting to different feature combinations and enhancing the fusion effect.

Additionally, the derivative of the Swish function changes smoothly, ensuring smoother gradient flow during training. This is particularly beneficial in deep networks, as Swish helps to mitigate gradient vanishing issues and improve overall training stability, as shown in Eqs. (7)–(9).

$$f(x) = \max(0, x). \tag{7}$$

$$\mathrm{Swish}(x) = x\sigma(\beta x). \tag{8}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \tag{9}$$

Among them, σ is the Sigmoid activation function, and $0 < \sigma < 1$ is a learnable parameter. Using parameter β to dynamically control the linear and nonlinear states of the activation function enables neurons to adaptively activate the appropriate state. This approach significantly reduces network parameters and markedly improves performance.

Fig. 6 illustrates the specific deployment of the Dynamic weighted feature fusion structure. In the bidirectional inter-layer fusion computation, both top-down and bottom-up, the spatial information in the feature maps that may contain safety helmet targets is enhanced, facilitating high-level feature maps to capture more complete features. This module addresses the issue of insufficient detection accuracy in safety helmet detection.

3.4. Lightweight network

When processing visual information, the human brain does not pay attention to every detail within the field of view, but prioritizes moving objects, prominent features, or familiar targets. This selective processing method reduces the computational load and energy consumption of the brain while still effectively completing tasks. In the field of safety helmet detection, traditional methods often conduct comprehensive and indiscriminate analysis of monitoring images, consuming a large amount of computing resources and time. Drawing on the efficient processing mechanism of the human brain, this paper proposes an innovative improvement scheme based on FasterNet, which brings significant performance improvements to helmet detection.

3.4.1. FasterNet block

FasterNet is an advanced neural network architecture. It has achieved faster operating speeds than other networks on various devices without affecting accuracy in various visual tasks. Partial Convolutional Blocks (PConv) in FasterNet optimize convolution operations by convolving only the valid regions, while not processing the invalid regions [40]. PConv significantly reduces the number of parameters and computational complexity by dynamically cropping the convolution

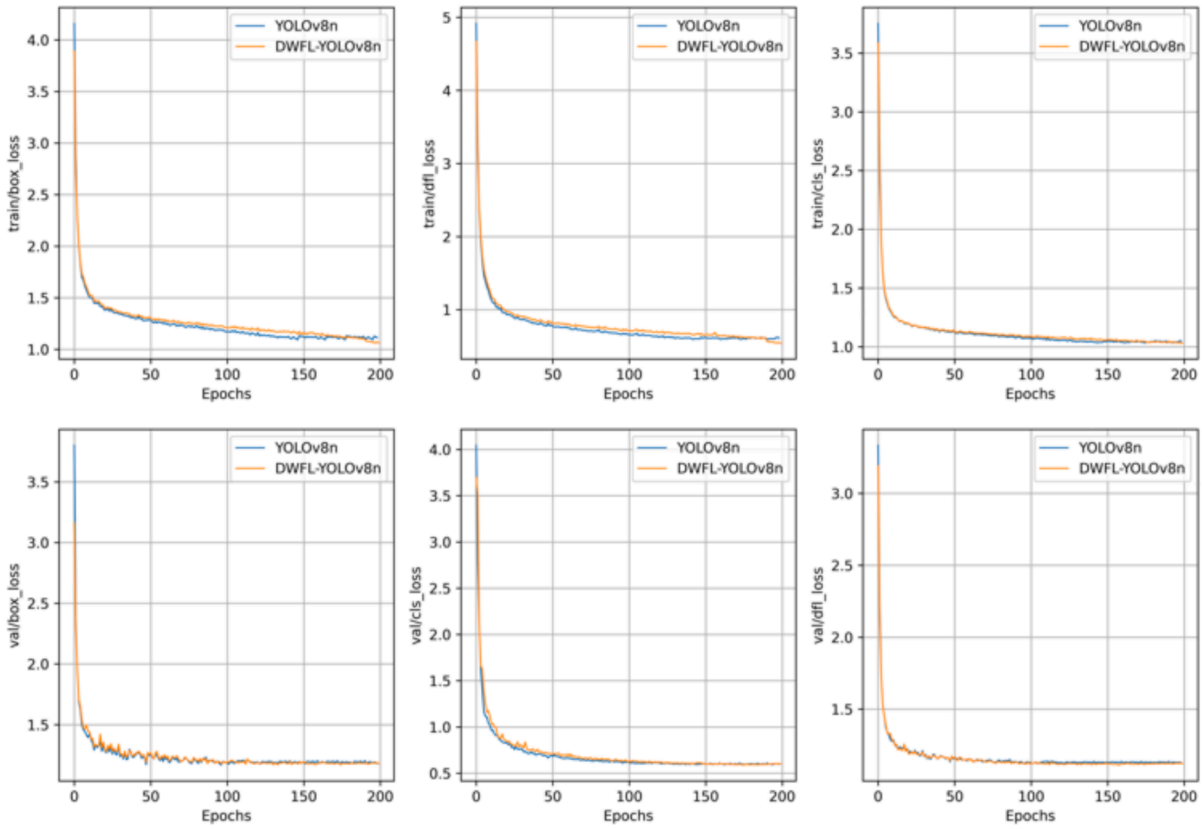


Fig. 9. Comparison of loss value curves.

calculation range. Compared to standard convolution, it avoids redundant calculations on invalid regions and is more suitable for the needs of lightweight networks. And by selecting effective regions, the processing of sparse feature maps can be more accurate, which can extract more accurate and effective features in helmet detection scenarios facing occlusion or feature loss [41].

As shown in Fig. 7, standard convolution operations involve convolving all input channels, with the number of channels in the convolutional kernel matching that of the input feature map, ensuring that each channel contributes to feature extraction. In Depthwise Convolution (DWConv), the convolution kernels are single-channel, meaning each kernel processes only one channel of the input feature map, producing an output feature map with the same number of channels as the input. In contrast, PConv functions as a division of labor mechanism, where certain channels focus on extracting spatial features while others are temporarily ignored. This strategy enhances global efficiency through targeted local optimization. The FLOPs for regular convolutions are:

$$h \times w \times k^2 \times c. \tag{10}$$

The FLOPs for DWConv are:

$$h \times w \times k^2 \times c^2. \tag{11}$$

In Eqs. (10) and (11), h represents the height of the output feature, w represents the width of the output feature, c represents the number of input channels, and k represents the size of the convolution kernel.

Reducing the number of parameters and computational overhead while ensuring effective utilization of all channel information and maintaining diversity in feature extraction are key objectives. The inclusion of normalization and activation layers further enhances the model's stability and performance. The FLOPs of PConv are shown in Eq. (12), where c_p is the number of partial input channels:

$$h \times w \times k^2 \times c_p^2. \tag{12}$$

3.4.2. Partial convolutional gated linear units

Drawing on the sensitivity of the human brain to salient features, we have designed a PCGLU module to enhance salient features. The safety helmet has unique prominent features such as shape and color, and the convolution kernel of PConv is specially designed for these features. PConv performs convolution only on a subset of channels, which may result in certain channel-specific local information being underutilized, especially when important features are distributed across channels that are not selected. To address this limitation, the Partial Convolution GLU (PCGLU) module combines Partial Convolution (PConv) with Gated Linear Units (GLU). The designed PCGLU module is used to replace the Bottleneck block in C2f, resulting in a new C2f PCGLU module to replace the C2f layer of the Neck network.

In the context of safety helmet detection, helmets are often located in specific regions of the image, while the surrounding background may be complex and contain considerable interference. The convolutional layers of PCGLU, coupled with the dynamic gating mechanism, provide precise feature selection in local feature modeling. This enables the model to effectively capture critical local features, such as the helmet's shape, color, edges, etc., while ensuring that these important local features are efficiently extracted. Local features are particularly crucial for helmet detection, as helmets are typically small objects, and details such as shape, color, and edges are key to recognition.

We introduce the FasterNet Block before the activation function of the GLU, transforming it into a partial convolution-based gated attention module. GLU consists of two linear projections, one of which is gated through an activation function. In PCGLU, the addition of partial convolution enables the module to dynamically adjust its receptive field based on the effective region of the input data. This adaptability enhances the module's robustness and efficiency, particularly in cases

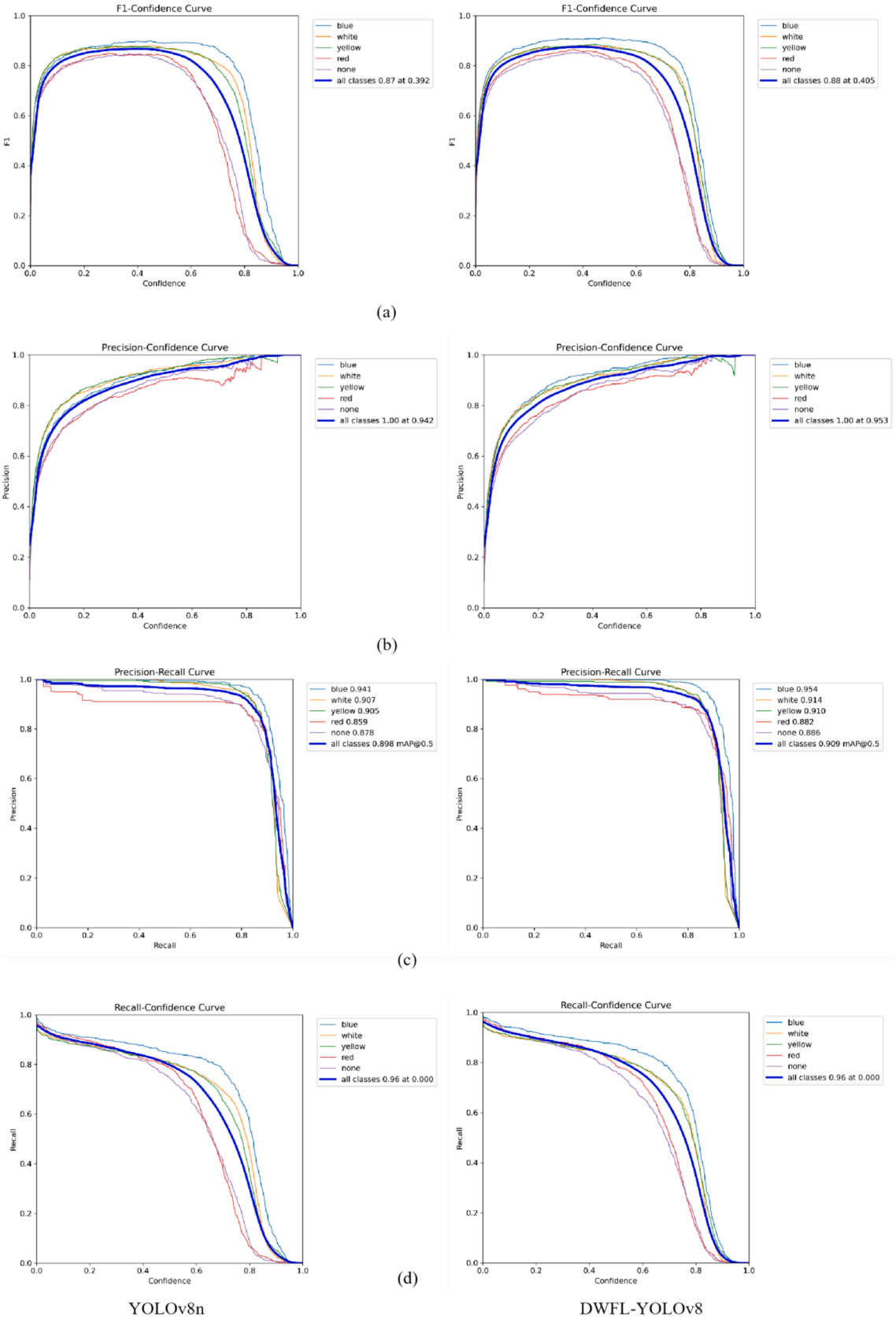


Fig. 10. Comparison of P curve, R curve, PR curve, F1 curve.

Fig. 11. The detection performance of DWFL-YOLO on the GDUT-HWD dataset.

involving incomplete or occluded inputs.

The partial convolution mechanism ensures that only relevant regions of the image are processed, reducing unnecessary computational overhead while focusing on geometric structures that matter most. The gating mechanism of the GLU then acts to selectively highlight the most important geometric features, allowing the network to effectively model the relationship between different local regions of the object and its environment. This combination ensures robust feature extraction, improving detection accuracy even in challenging conditions like occlusion or complex backgrounds.

As shown in the Fig. 8, both the GLU module and the improved PCGLU module are composed of three key components: (1) a gating mechanism that enhances the extraction of critical information, (2) partial convolutions for local feature extraction, and (3) a fusion and residual connection mechanism that strengthens feature integration and information flow.

First, the input feature y is mapped into two parts x, v through a linear operation. This linear transformation mainly involves generating the gating signal, which determines the activation level of each channel. Let the weight matrix of the linear transformation be $W_{gating} \in \mathbb{R}^{C\times C}$ (corresponding to the linear transformation for each channel), and the bias term be $b_{gating} \in \mathbb{R}^{C}$. Then, the transformation can be expressed as Eq. (13):

$$G = \mathrm{Linear}(\mathrm{y}) = w_{gating}\cdot y + b_{gating}. \tag{13}$$

The value branch x performs convolution calculation in the effective area through partial convolution, as shown in **Eq.** (14):

$$\mathrm{x}' = \mathrm{Act}(PartialConv(x)). \tag{14}$$

The gating operation combines the value branch with the gating branch through point wise multiplication, as shown in **Eq.** (15):

$$\mathrm{x}'' = x' \odot v. \tag{15}$$

Finally, the output feature is obtained by adding the residual of x '' and the input feature x shortcut, as shown in **Eq.** (16):

$$\mathrm{y} = x_{shortcut} + \mathrm{Linear}(x''). \tag{16}$$

4. Experimental results and discussion

This section evaluates the detection performance of the proposed model on three public available safety helmet dataset and compares it with existing conventional object detection algorithms to verify the feasibility of the proposed algorithm.

4.1. Implementation details

The experiment was conducted on an NVIDIA GeForce 3080 Ti (12 GB) GPU with Python 2.1.0, Python 3.10, and CUDA 12.1 dependencies. The input image size was set to 640 x 640, with a batch size of 16 for the training data. the models utilized in all experiments were trained on the training set for 200 epochs, the initial learning rate was set to 0.01, with an optimizer weight decay of 0.0005 and a preheating initial momentum of 0.937.

4.2. Datasets

This research utilized the publicly available GDUT Hardhat Wear Detection (GDUT-HWD) dataset [42], which contains 3,174 images. Of

Fig. 12. The detection effect of DWFL-YOLO on GDUT-HWD dataset.

these, 2,177 images were allocated for training, while the remaining 997 were used for testing, following an 8:2 split. In total, the dataset contains 18,893 annotated instances. As presented in Table 1, the dataset consists of five categories for detection: blue, white, yellow, red helmets, and no helmet. Each instance is labeled with a class and corresponding bounding box. The GDUT-HWD dataset covers diverse and complex environments, including different scenes, lighting variations, personal postures, and instances of occlusion.

Moreover, the dataset classifies helmets into three size categories based on area: small (less than 322 pixels), which makes up 47.4 % of the dataset; medium (between 322 and 962 pixels), comprising 41.9 %; and large (greater than 962 pixels), accounting for 10.7 %. This size-based classification adds an extra layer of difficulty for detecting helmet use, particularly in small-object detection, requiring the model to perform well in more challenging environments and with more refined classification criteria.

The SHWD dataset is designed for helmet use and head detection. It includes 7820 safety helmet images and facial images of targets at different scales, capturing diverse practical scenarios, such as construction sites and production plants. This dataset contains a total of 11,345 annotated instances, including 763 instances of helmets and 10,582 instances of person. In the experiment, the dataset was split into training set, validation set, and testing set in a ratio of 8:1:1.

The CUMT-HelmeT dataset [43] was released by the Intelligent Detection and Pattern Recognition Research Center of China University of Mining and Technology in 2023. The dataset contains 6421 images, labeled using Labelimg software. The dataset is mainly used for detecting the wearing of safety helmets in complex and dimly lit underground environments. It was split into training, validation, and testing sets in a ratio of 8:1:1.

4.3. *Performance evaluation of lightweight model*

The DWFL-YOLO model proposed in this article is compared with the benchmark model YOLOv8n and several existing classic single-stage object detection algorithms, including SSD, Faster R-CNN, YOLOv5s, YOLOv5n, YOLOv7-Tiny [44], YOLOX-Tiny [45], YOLOv8s, YOLOv10n, YOLOv11s and YOLOv11n. The comparison focuses on key metrics such as model parameters, FLOPs, model size, and detection accuracy. FLOPs are utilized to measure the computational complexity of each model.

To further demonstrate the performance of the DWFL-YOLO model and quantitatively analyze the detection performance of the proposed algorithm compared to other algorithms, we attempted to validate the model on different datasets. Comparative experiments were conducted on the GDUT-HWD dataset, comparing our algorithm with SSD, Faster R-CNN, RT-DETR, YOLOv5s, YOLOv5n, YOLOv7 Tiny, YOLOX-Tiny, YOLOv8s, YOLOv8n, YOLOv10n, YOLOv11s and YOLOv11n. Table 2 lists the values of model parameters, complexity, model size, and average accuracy for different algorithms. From the table, it can be seen that our algorithm exhibits the highest accuracy. In addition, compared with the YOLOv8n benchmark algorithm, the parameters, FLOPs, and model size were reduced by 34.9 %, 14.8 %, and 1.6 % respectively, and the mAP (0.5) was improved by 1.2 %. Compared with the latest YOLOv11n, the number of parameters has decreased by 24 % and FLOP has decreased by 8.7 %. The model size is only slightly larger than YOLOv5n, YOLOv10n and YOLOv11n.

Table 3 presents the experimental results of several state-of-the-art algorithms on the SHWD dataset. The comparison of results demonstrates that the proposed algorithm achieves a mAP of 93.8 %, indicating an improvement in accuracy compared to other algorithms. In Table 4, the superiority of the DWFL-YOLO algorithm is further validated on the CUMT-HelmeT dataset. The experimental results indicate that,

Fig. 13. The detection effect of DWFL-YOLO on SHWD dataset.

compared to SSD, Faster-RCNN, RT-DETR, YOLOv5n, YOLOv8n, YOLOv10n and YOLOv11n, the mAP increases by 5.2 %, 11.7 %, 1.5 %, 2.6 %, 1.3 %, 0.6 % and 1.1 %, respectively, thereby confirming the superiority of the DWFL-YOLO algorithm.

4.4. Ablation experiment

To evaluate the detection performance of our proposed DWFL-YOLO algorithm, we conducted a series of ablation experiments.

In Table 5, "DWFF" stand for dynamic weighted feature fusion structure. Method (1) indicates that using only the diverse branch block improves accuracy without changing the number of model parameters. Method (2) shows that when the neck network uses the DWFF architecture, the accuracy remains almost the same compared to YOLOv8n, but the number of model parameters is reduced by 32.2 %. Method (3) demonstrates that introducing both DBB module and DWFF structures keeps the number of model parameters the same as in method (2), resulting in a 0.3 % increase in mAP. Method (4) introduces DWFF module and PCGLU module, further reducing the number of parameters and computation on the basis of the previous one. The model structure selection of method (5) ensures the optimal performance of DWFL-YOLO. DWFL-YOLO ultimately achieved 90.9 % mAP and 1.96 M parameters on the GDUT-HWD dataset. Compared with the YOLOv8n baseline, mAP increased by 1.2 %, parameters decreased by 34.8 %, and computational complexity decreased by 14.8 %. The experimental results verify the effectiveness of the various improvement measures.

Table 6 compares a variety of advanced lightweight target detection models. From the table data, we can see that the improved DWFL-YOLO algorithm in this paper is the lowest in terms of model parameters. FLOPs is only higher than both the YOLOv8 GhostHGNetV2 algorithm and the EfficientDet algorithm. It performs well in maintaining lightweight characteristics, has the advantages of fewer parameters, low computation, and small model size. At the same time, it surpasses other comparison models in detection accuracy, which means that it can reduce computing resource consumption, improve operating efficiency, and ensure high detection accuracy in practical applications.

Table 7 shows the experimental results of the DBB model on different datasets, comparing the performance of two models, YOLOv8n and YOLOv8n-DBB, on three datasets: GDUT-HWD, SHWD, CUMT-Helmet and VOCdevkit2012. The evaluation metrics include mAP(0.5), mAP (0.5–0.95), and Recall (R). The results show that YOLOv8n-DBB performs better than or close to YOLOv8n on most datasets and metrics, especially having a more significant advantage on the CUMT-Helmet dataset.

In Table 8 and Table 9 compare the performance of the BiFPN and the improved DWFF structures on the GDUT-HWD dataset, as well as the performance of the DWFF module using different activation functions. These results indicate that introducing the DWFF structure into lightweight networks significantly improves the detection ability of safety helmets.

Table 10 shows that using the improved PCGLU module ensures lightweight models while improving accuracy, proving the effectiveness of the module.

The detection performance of the proposed DWFL-YOLO model has been evaluated against the benchmark model YOLOv8n, as well as several state-of-the-art lightweight single-stage target detection algorithms, such as SSD, YOLOv5s, YOLOv7-Tiny, YOLOX-Tiny, and YOLOv8s. The key performance metrics include the average precision

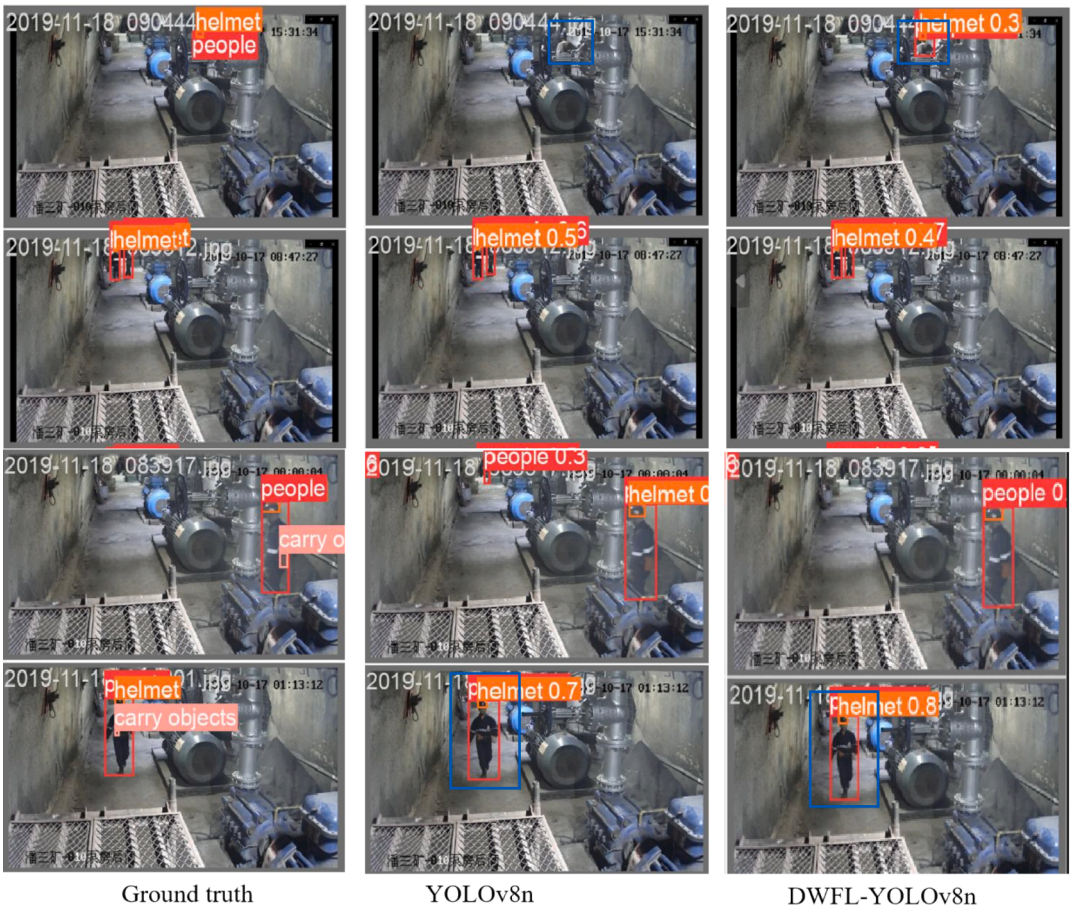


Fig. 14. The detection effect of DWFL-YOLO on CUMT-HelmeT dataset.

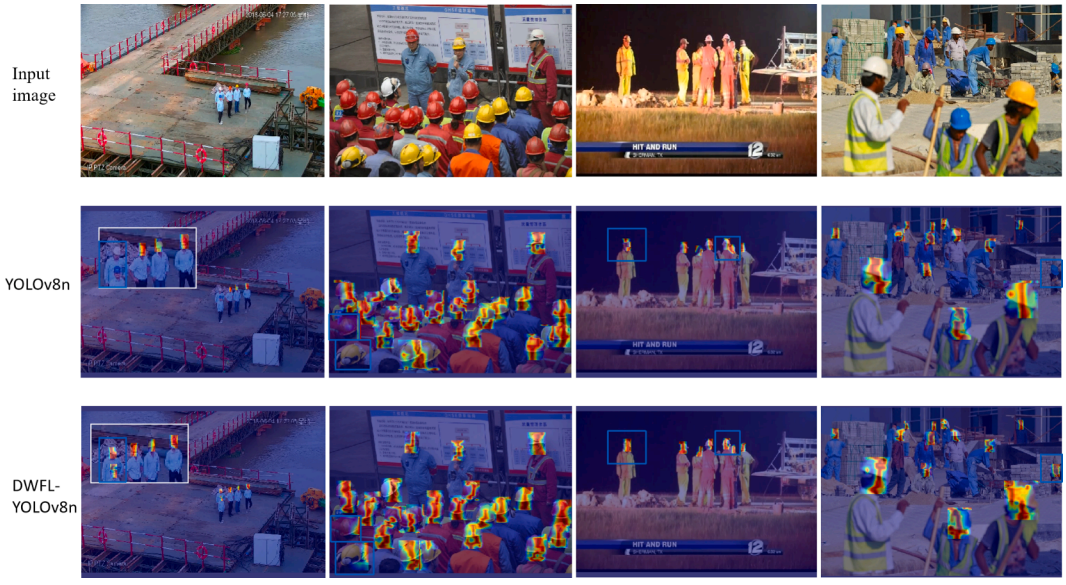
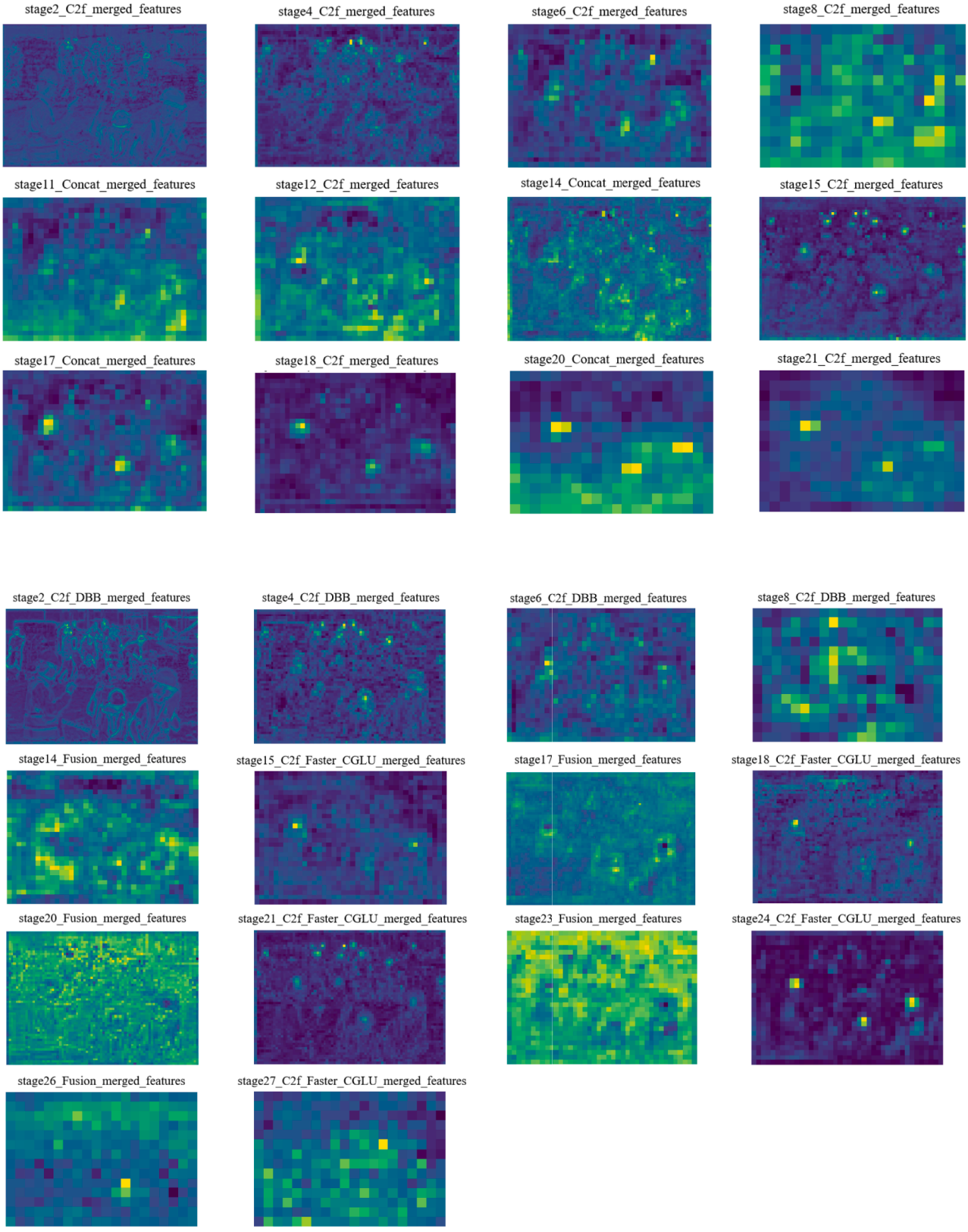


Fig. 15. Comparison of heat maps.

(AP) for each class and the mean average precision (mAP) across all classes, both of which are essential for assessing object detection effectiveness.

Fig. 9 compares the various loss curves of YOLOv8n and DWFL-YOLO training and verification. It can be seen that each loss curve of DWFL-YOLO is lower than the former, the fitting is excellent during training, and the generalization is strong during verification, showing better performance. As shown in Fig. 10, this study performs a multi-dimensional performance comparison between YOLOv8n and the improved YOLOv8n, including F1-confidence curves, precision-confidence curves, recall-confidence curves, and precision-recall curves. From the comparison results, it is clear that the improved



(b) Feature maps of DWFL-YOLOv8n

Fig. 16. Comparison of feature maps.

YOLOv8n demonstrates significant advantages: In the F1-confidence curve, the improved model's curve is higher in most confidence intervals, indicating better overall precision and recall performance; the precision-confidence curve shows that the improved model has higher precision in high-confidence areas, meaning its predictions are more reliable; in the precision-recall curve, the improved model is closer to the top-right corner, balancing higher precision and recall; in the recall-confidence curve, the improved model's curve drops more gently, indicating that it can still effectively recall targets at high confidence. These results strongly demonstrate that the improved YOLOv8n has significant improvements in both precision and recall capabilities for object detection. compares the actual measured values of confidence, accuracy, recall and F1 score of YOLOv8n and DWFL-YOLO algorithms. Different color curves represent different categories, as well as summary indicators of all categories. The overall curve position and final performance of DWFL − YOLOv8 are better, indicating that the algorithm is better in comprehensive consideration of precision and recall. Fig. 11 visualizes the detection performance on the GDUT-HWD dataset. These results demonstrate that the proposed approach maintains robust detection performance within a lightweight network structure.

As shown in Fig. 12, the detection results of the baseline model YOLOv8n and DWFL-YOLO for different color identification targets are compared on the GDUT-HWD dataset. YOLOv8n algorithm can detect some target colors, but the confidence level is low, such as 0.3 for red target detection; The color category judgment of some targets is also inaccurate, and some targets may not be completely detected. Compared with YOLOv8n, the confidence level of DWFL-YOLO detection is improved, for example, the confidence level of some yellow target detection is increased from 0.8 to 0.9; The color category of the target is judged more accurately, for example, the "blue" category can be detected, and the detected target is more complete.

Fig. 13 shows the detection results of DWFL-YOLO on the SHWD dataset. The detection targets are mainly people and hats. DWFL-YOLO has improved the confidence level of people and hat detection. For example, for the detection of the same hat, the confidence level has increased from 0.6 to 0.7; The detection effect is closer to the real annotation. Fig. 14 shows the comparison object detection results of CUMT Helmet dataset. The selected original image brings challenges, such as small object detection, multi object detection and poor lighting conditions. Compared with YOLOv8n, DWFL − YOLOv8n has significantly improved the confidence of detection, such as the confidence of helmet detection from 0.53 to 0.89; And it performs better on the integrity of target detection, and can detect targets in the scene more comprehensively.

To more intuitively demonstrate the comparison of detection effects before and after the improvement of the model in this paper, Fig. 15 presents a comparison of the visualization results of heat maps between the improved algorithm in this paper and the YOLOv8n algorithm. The improved algorithm can focus better on the area where the safety helmets are located. This indicates that the above mentioned improvements, while reducing the model complexity, make the extraction of feature maps more focused on the objects of interest, suppress the interference from the background, and improve the accuracy.

Fig. 16 compares the feature map of the improved layer with the feature map of the corresponding layer of the original YOLOv8n. The number of channels in each layer is fused to generate a feature map. The comparison results show that the improved YOLOv8n has significant advantages in feature representation and utilization. Although the original YOLOv8n can perform basic feature extraction and fusion tasks, it is relatively weak in focusing on target features and capturing spatial information. In contrast, the improved version introduces new modules and fusion methods, allowing it to highlight target features more effectively, reduce background noise interference, and enhance the feature map's capability in representing targets.

From the above comparison results, it can be seen that the DWFL-YOLO algorithm has multiple advantages in object detection. The confidence level for detecting objects such as people, helmets, and color − marked items has been significantly improved, enabling more reliable identification of detected objects. It reduces the situation of object missed detection. In different scenarios, it can more comprehensively detect all objects. The classification judgment of objects (such as color categories) is more accurate, reducing classification errors. The positioning of objects is more precise, and the heat distribution is more in line with the actual positions of objects.

Overall, the DWFL-YOLO algorithm has improved in terms of the accuracy, completeness, and precision of object detection, and can better meet the needs of practical applications.

5. Conclusion

This paper declares the introduction of a lightweight helmet detection algorithm, DWFL-YOLO, based on weighted multi-scale feature fusion. The algorithm applies bidirectional dynamic feature fusion and lightweight network modules to balance the accuracy and ease with which lightweight networks and algorithms can be implemented. Firstly, diverse branch blocks replace the C2f layer, enabling the backbone network to effectively extract multi-scale features. Secondly, a bidirectional dynamic feature fusion network was designed based on the BiFPN structure, which enhanced feature extraction, reduced model parameters, and improved detection accuracy. Finally, the PCGLU module was introduced, which not only reduces unnecessary calculations but also highlights key features, achieving a lightweight model. The experimental results on the GDUT-HWD dataset show that this method has good detection performance for helmet detection in complex backgrounds, fully meeting requirements for safety helmet detection in normal daytime ambient light settings. However, in practical applications, numerous practical scenarios frequently involve unstable or completely insufficient lighting conditions. For example, in industrial sites, construction sites, or public safety areas, low-light conditions pose significant challenges to the accuracy and robustness of object detection algorithms. Therefore, future studies should place the response of the model's detection capabilities under test in nighttime or light-deficient contexts.

CRediT authorship contribution statement

Hongge Ren: Writing – review & editing, Supervision, Project administration, Funding acquisition. **Anni Fan:** Writing – original draft, Software, Methodology, Conceptualization. **Jian Zhao:** Supervision, Resources, Project administration. **Hairui Song:** Visualization, Validation, Software. **Zhiguo Wen:** Validation, Investigation. **Shuai Lu:** Validation, Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hongge Ren reports financial support was provided by Tianjin Municipal Science and Technology Commission. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. The publicly available

dataset utilized in this research can be accessed via the following link: https://github.com/wujixiu/helmet-detection.

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